



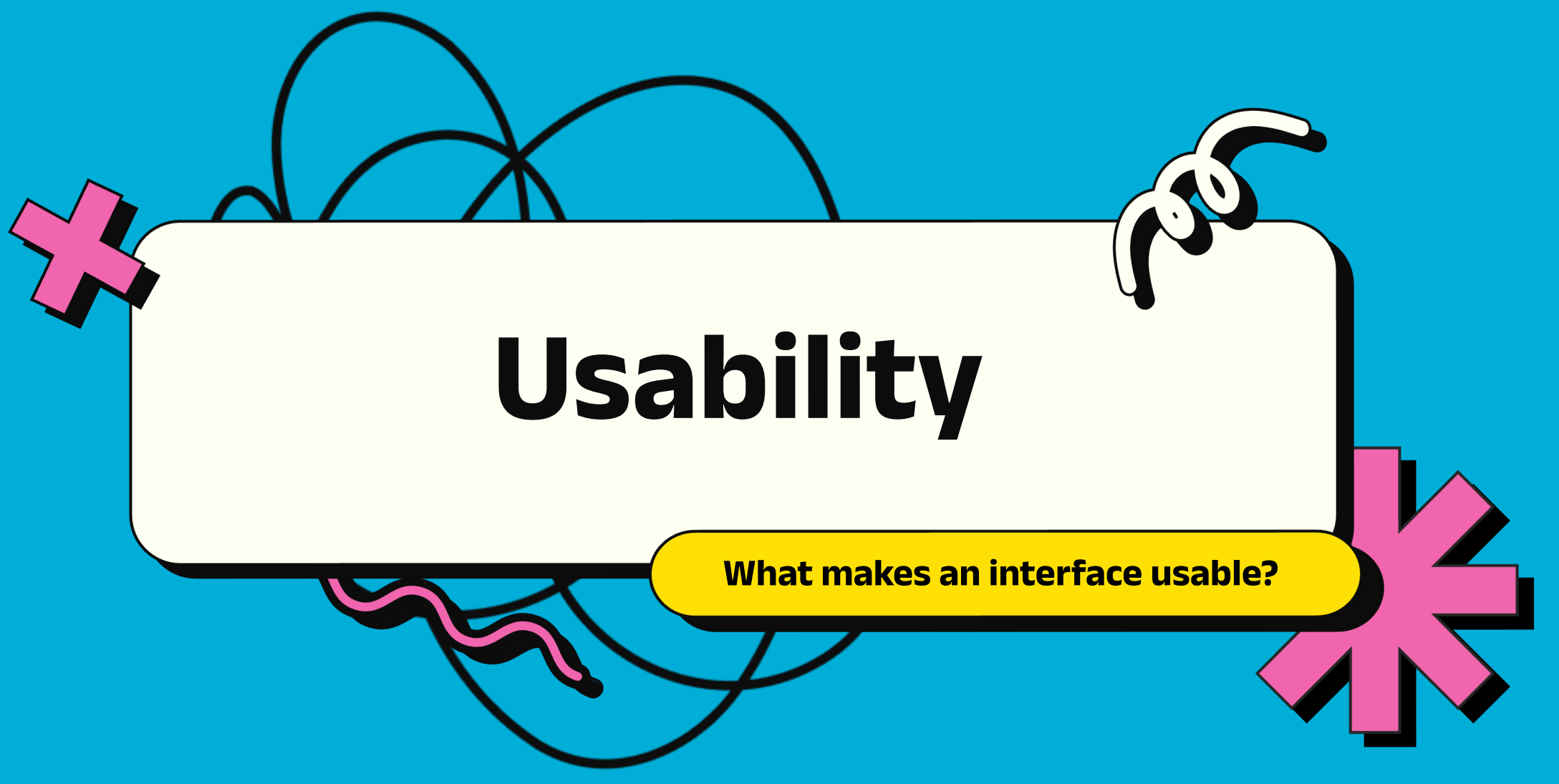
Interface Design

IDE2IL



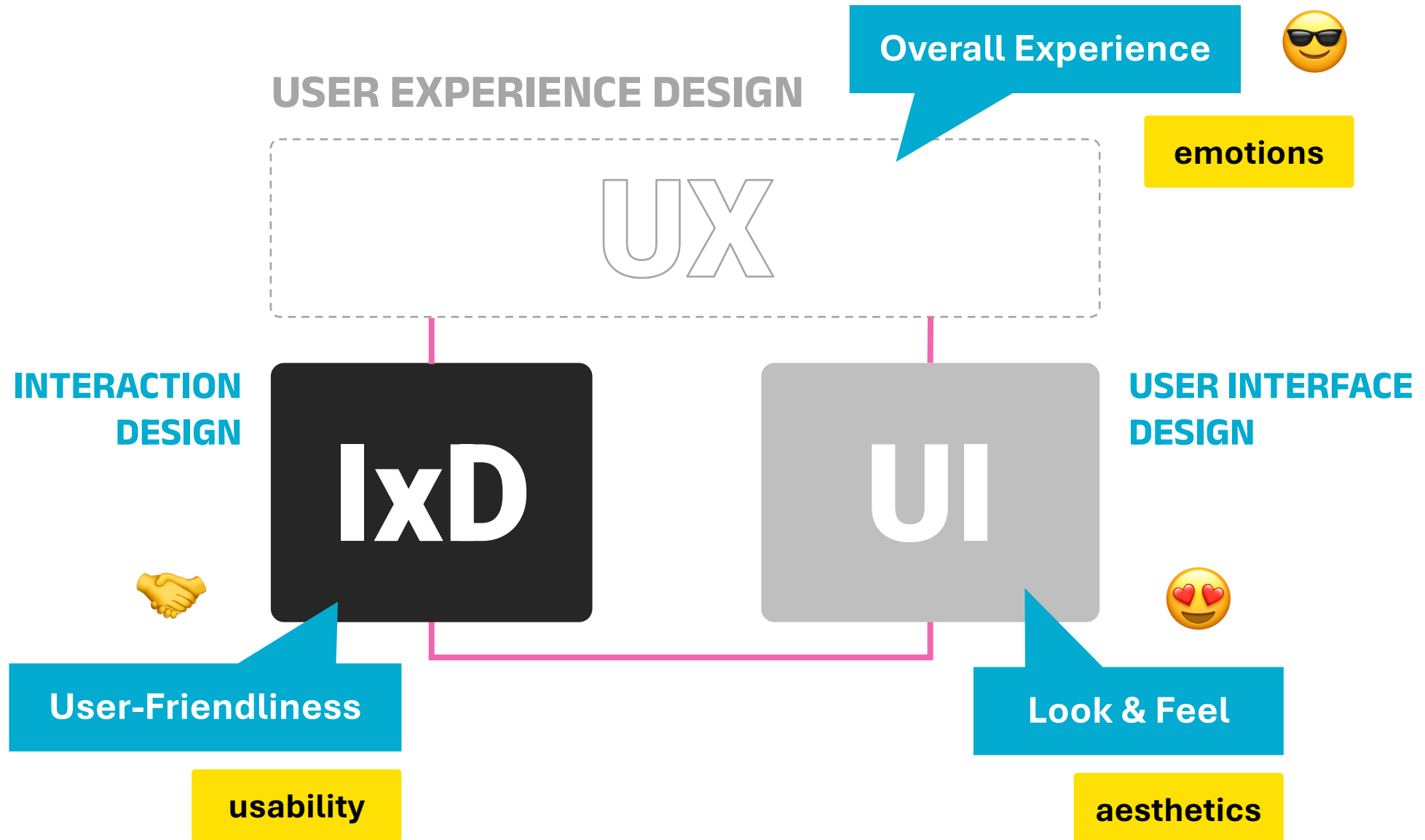
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Usability

What makes an interface usable?



ANTICIPATED USE

Perception about a product without having used it

ACTUAL USE

Effective, efficient, satisfactory task completion

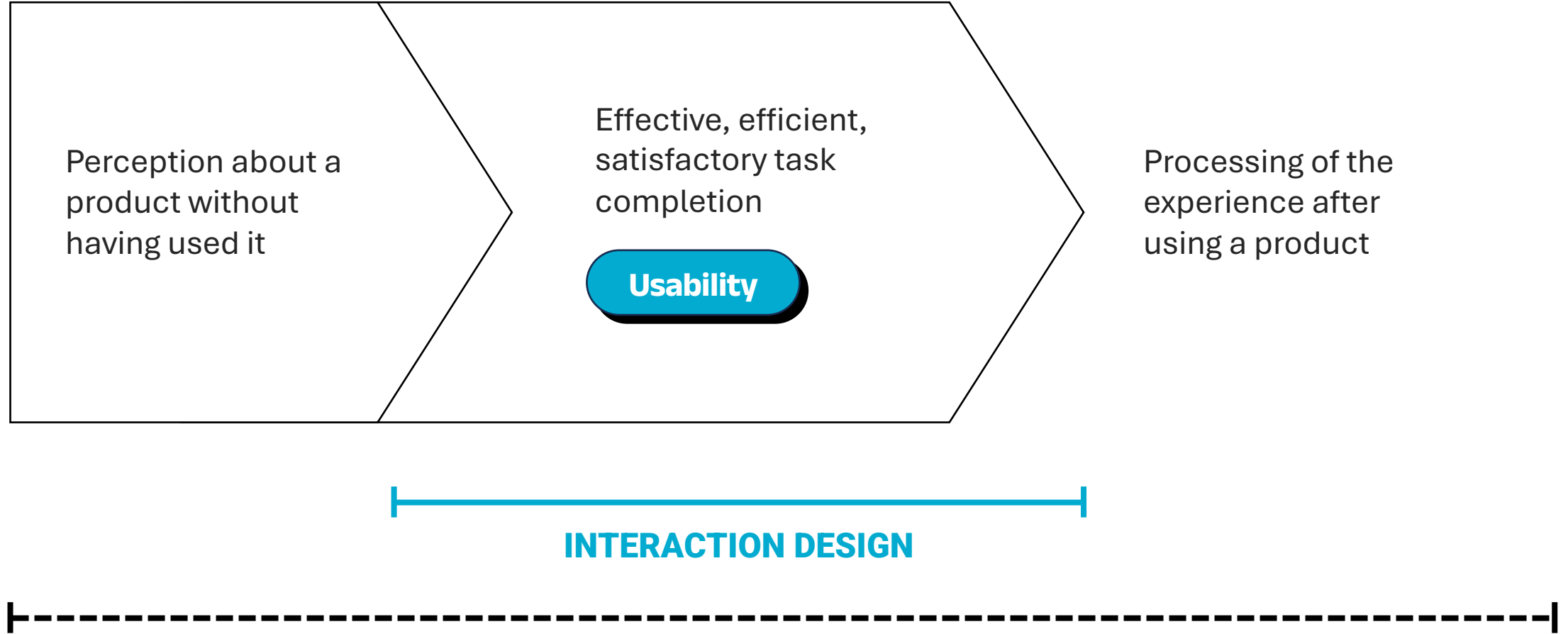
Usability

DIGESTED USE

Processing of the experience after using a product

INTERACTION DESIGN

USER EXPERIENCE DESIGN



Effectiveness / Efficiency

- **Effectiveness:**

How good an interface facilitates is in accomplishing the task for which it was intended

>> *success rate, error rate, completion rate...*

- **Efficiency:**

The rate or speed at which an interface enables a user to accurately and successfully complete a task

>> *completion time, response time, number of clicks...*

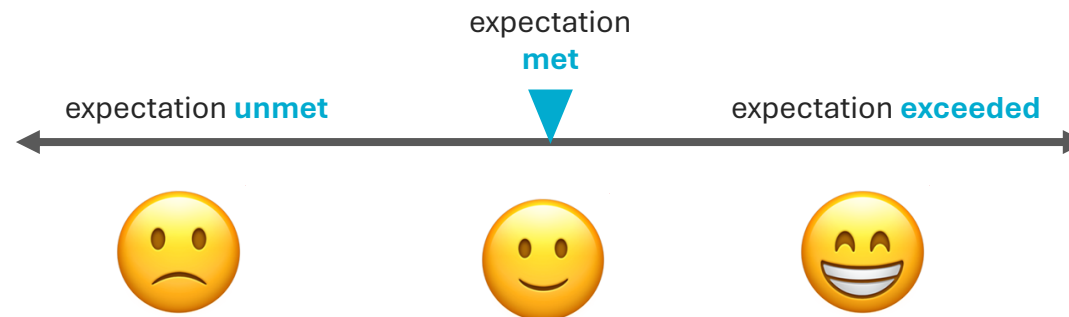


Satisfaction

Satisfaction refers to extent to which the user's physical, cognitive and emotional responses that result from the use of a system, product or service meet the user's needs and expectations.

It may be measured by openly asking users for their opinions, or via questionnaires. They may cover diverse factors such as:

- ease-of-use
- mental demand
- physical demand
- temporal demand
- effort
- ...

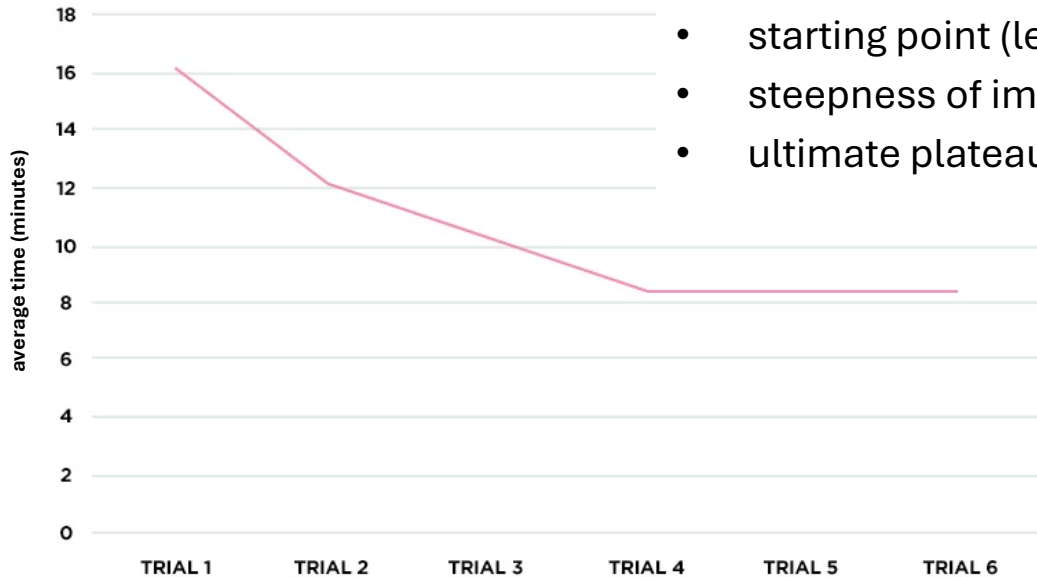


Learnability / Memorability

- **Learnability:** How easy it is to accomplish a task the first time they encounter an interface
- **Memorability:** How easy an interface is to remember how to use, once learned

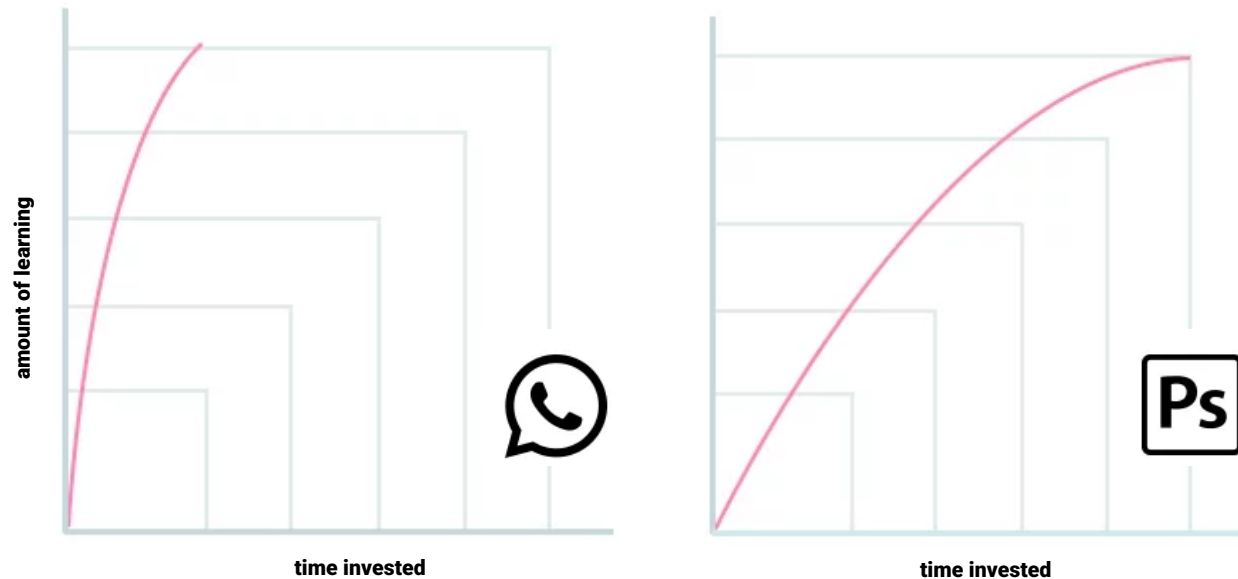
Measured in terms of learning curves:

- starting point (learnability)
- steepness of improvement (memorability)
- ultimate plateau (efficiency)



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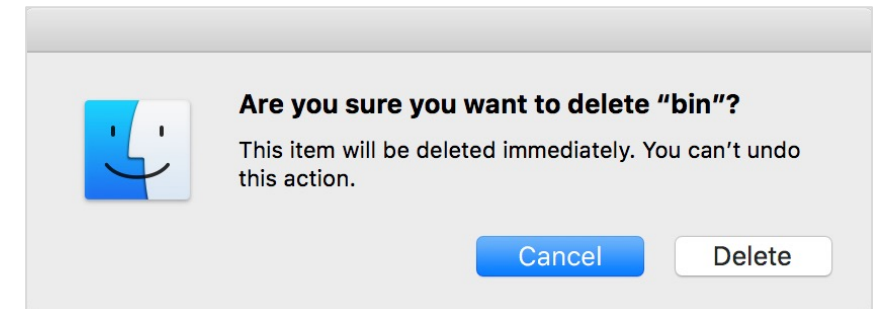
Depends on:

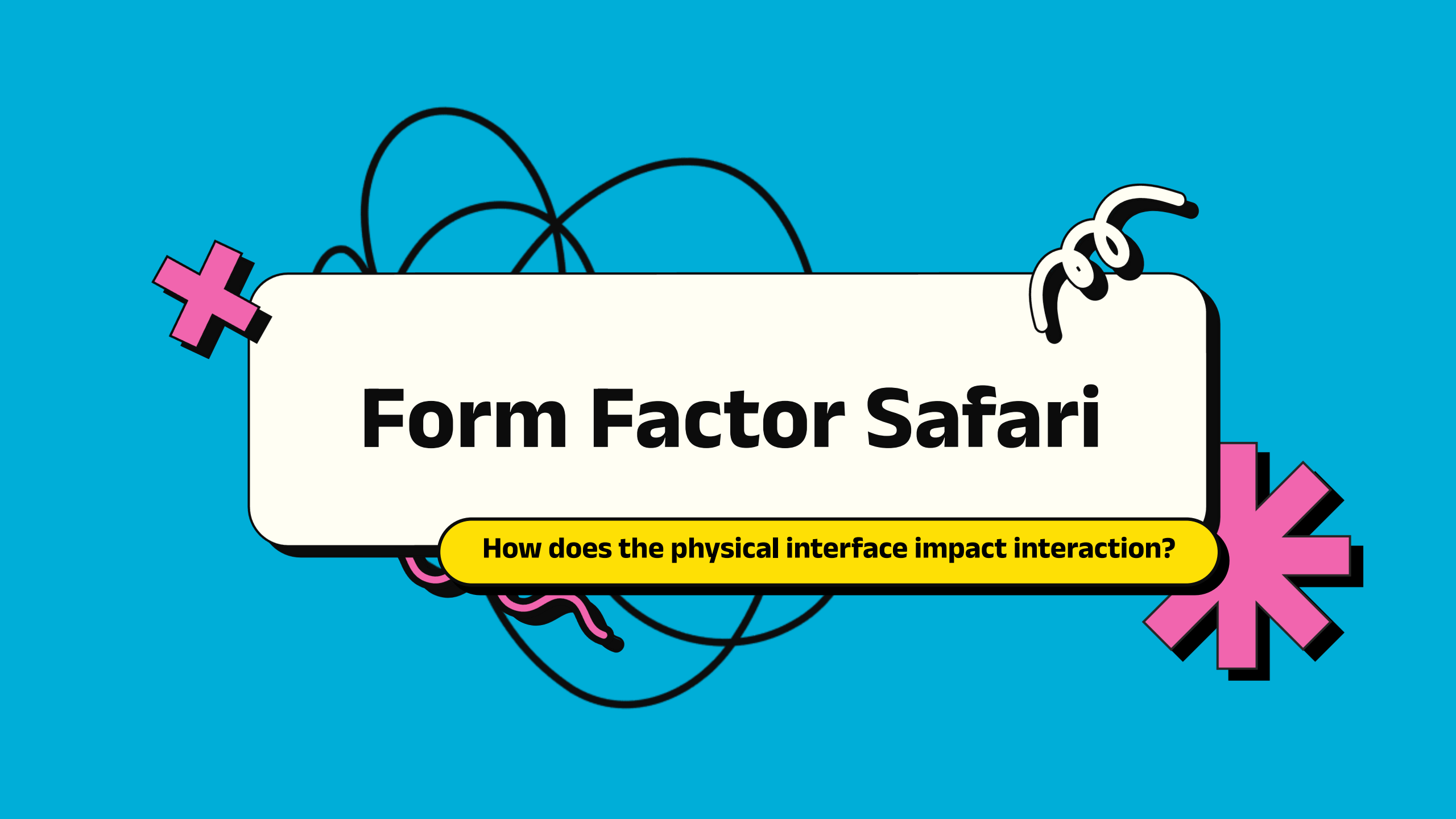
- complexity of the product (e.g., number of features)
- user's pre-existing knowledge
- clarity of the user interface
- guidance (e.g., onboarding)
- ...

Safety

Safety is concerned with how well users are protected from dangerous conditions and undesirable situations:

-  **ergonomic**
external conditions (e.g., hazardous environment)
-  **cognitive**
risk of carrying out unwanted actions (e.g., pressing the wrong button)
-  **emotional**
fears of the consequences of making errors (e.g., loss of data)





Form Factor Safari

How does the physical interface impact interaction?



	smartwatch	smartphone	tablet	desktop	public display
screen	XS	S	M	L	XL
main input	touch	touch	touch	mouse, trackpad	controller, touch, pen
orientation	portrait (shape!)	portrait (landscape)	portrait, landscape	landscape	landscape
additional input	voice, acceleration, location	cameras, acceleration/orientation, location, voice	cameras, acceleration/orientation, location, voice	camera (voice)	camera (voice)
output	visual, tactile	visual, tactile, audio	visual, tactile, audio	visual, audio	visual, audio
connectivity, usage time, static vs. non-static environment, indoor vs. outdoor environment...					

>> different device groups provide different services in different contexts

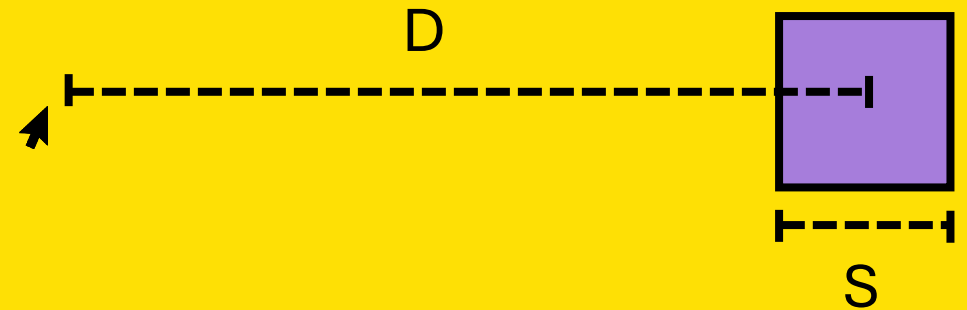
FITTS's



LAW

preview:
IM master's

The time to acquire a target is a function of distance to the target (D), and size of the target (S).



<http://simonwallner.at/ext/fitts/>

- Paul M. Fitts

Desktop & Mouse Interaction

Hover me I'm awesome

The traditional way of computing happens in front of a screen, with mouse and keyboard.

- **relatively large screens (L)**
in a static, focused environment
- **high input precision**
mouse cursor gives us **1-pixel precision**
- **hover states**
system knows where users are looking before they click,
>> hover animations, tooltips, revealing hidden menus
- **keyboard synergy**
interaction is often two-handed,
>> shortcuts and modifier keys to bypass UI menus
- **touchpad synergy on laptops**

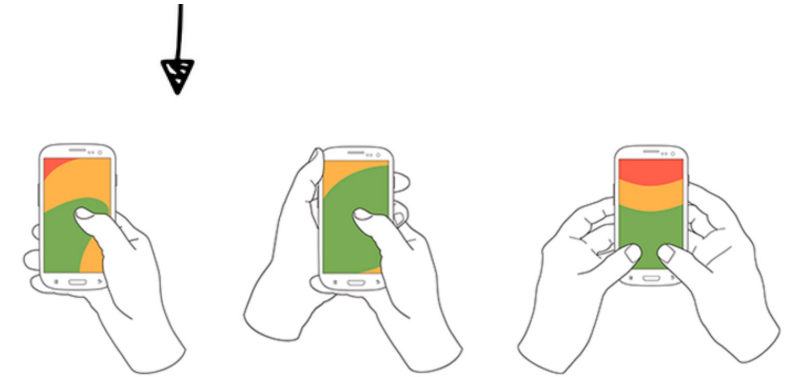


Mobile & Touch Interaction

Today's most commonly used computers are portable and touch-first.

- **relatively small screens (XS-M)**
in variable, non-static, distracting environments
- **low input precision**
pointer (finger) occludes the screen ("**fat finger problem**")
>> requires large touch targets and ample spacing
- **reachability & ergonomics**
depends heavily on device size and grip
>> primary actions must live in comfortable "thumb zones"
- **sensor synergy**
devices are packed with all sorts of sensors
>> interaction via gestures, voice, camera, location...

"thumb zone"



"fat finger problem"



Consoles & Controllers

Spatial computing driven by tactile hardware, from "lean-back" TVs to handhelds.

- **flexible size**
from TVs in living rooms to portable handhelds
- **tactile input controller**
with thumbsticks, d-pads and physical buttons
>> instant haptic feedback without looking, muscle memory
- **focus states**
are required in discrete UI navigation tasks (e.g., menus)
to signal users where they are
- **spatial & sensor synergy**
hardware packed with gyroscopes etc
>> enables motion aiming and immersive haptic feedback



Large Screen & Kiosk Interaction

Macro-level direct manipulation in public, shared, or stressful physical spaces.

- **extra-large screens (XL)**
in "step-up" environments like train stations, museums, or meeting rooms
- **reachability**
direct touch on massive screens causes whole-arm strain ("gorilla arm")
>> UI must localize interactions so users don't have to step back and forth
- **environmental sensor synergy**
hardware integrated into the physical architecture of a room
e.g., presence sensors to "wake up" on approach
e.g., NFC/RFID to bridge the physical/digital gap



Challenge #1

<https://elearning.fh-ooe.at/mod/assign/view.php?id=523231>



Fame
OR
Shame





Interface Design

IDE2IL

up next: IDE2IL 17.03.



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